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United States Patent	12396771
Kind Code	B2
Date of Patent	August 26, 2025
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Intra-osseous plate system and method

Abstract

An intra-osseous support structure can be used to fixate opposed portions of bone. In some examples, the intra-osseous support structure is positioned in openings formed in adjacent portions of bones. Fasteners are inserted through the bone portions to secure the intra-osseous support structure in the bones. Depending on the application, one or more external bone plates may also be applied to the bone portions. The external bone plate may be in compression while the intra-osseous support structure is in tension under load in situ.

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Appl. No.: 18/649799

Filed: April 29, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240277389 A1	Aug. 22, 2024

Related U.S. Application Data

continuation parent-doc US 17897321 20220829 US 11969193 child-doc US 18649799
continuation parent-doc US 16877159 20200518 US 11426219 20220830 child-doc US 17897321
continuation parent-doc US 15148774 20160506 US 10653467 20200519 child-doc US 16877159
us-provisional-application US 62157561 20150506

Publication Classification

Int. Cl.: **A61B17/80** (20060101); **A61B17/68** (20060101); **A61B17/86** (20060101); **A61B17/88** (20060101); A61B17/064 (20060101)

U.S. Cl.:

CPC **A61B17/8061** (20130101); **A61B17/686** (20130101); **A61B17/8014** (20130101); **A61B17/8095** (20130101); **A61B17/8872** (20130101); A61B17/0642 (20130101); A61B17/86 (20130101)

Field of Classification Search

CPC: A61B (17/8061); A61B (17/8071); A61B (17/8095)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2133859	12/1937	Hawley	N/A	N/A
2614559	12/1951	Livingston	N/A	N/A
2825329	12/1957	Caesar	N/A	N/A
3709218	12/1972	Halloran	N/A	N/A
4159716	12/1978	Borchers	N/A	N/A
4187840	12/1979	Watanabe	N/A	N/A
4338927	12/1981	Volkov et al.	N/A	N/A
4570624	12/1985	Wu	N/A	N/A
4627425	12/1985	Reese	N/A	N/A
4628919	12/1985	Clyburn	N/A	N/A
4754746	12/1987	Cox	N/A	N/A
4757810	12/1987	Reese	N/A	N/A
4790302	12/1987	Colwill et al.	N/A	N/A
4895141	12/1989	Koeneman et al.	N/A	N/A
4952214	12/1989	Comparetto	N/A	N/A
4959065	12/1989	Arnett et al.	N/A	N/A
4978347	12/1989	Ilizarov	N/A	N/A
4988349	12/1990	Pennig	N/A	N/A
5021056	12/1990	Hofmann et al.	N/A	N/A
5112334	12/1991	Alchermes et al.	N/A	N/A
5207676	12/1992	Canadell et al.	N/A	N/A
5254119	12/1992	Schreiber	N/A	N/A
5312412	12/1993	Whipple	N/A	N/A
5358504	12/1993	Paley et al.	N/A	N/A
5364402	12/1993	Mumme et al.	N/A	N/A
5413579	12/1994	Tom Du Toit	N/A	N/A
5417694	12/1994	Marik et al.	N/A	N/A
5439381	12/1994	Cohen	N/A	N/A
5449360	12/1994	Schreiber	N/A	N/A
5529075	12/1995	Clark	N/A	N/A

5586564	12/1995	Barrett et al.	N/A	N/A
5620442	12/1996	Bailey et al.	N/A	N/A
H1706	12/1997	Mason	N/A	N/A
5788695	12/1997	Richardson	N/A	N/A
5803924	12/1997	Oni et al.	N/A	N/A
5810822	12/1997	Mortier	N/A	N/A
5893553	12/1998	Pinkous	N/A	N/A
5935128	12/1998	Carter et al.	N/A	N/A
5941877	12/1998	Viegas et al.	N/A	N/A
5951556	12/1998	Faccioli et al.	N/A	N/A
6030391	12/1999	Brainard et al.	N/A	N/A
6162223	12/1999	Orsak et al.	N/A	N/A
6171309	12/2000	Huebner	N/A	N/A
6719773	12/2003	Boucher et al.	N/A	N/A
6743233	12/2003	Baldwin et al.	N/A	N/A
7001388	12/2005	Orbay et al.	N/A	N/A
7033361	12/2005	Collazo	N/A	N/A
7037309	12/2005	Weil et al.	N/A	N/A
7037342	12/2005	Nilsson et al.	N/A	N/A
7097647	12/2005	Segler et al.	N/A	N/A
7153310	12/2005	Ralph et al.	N/A	N/A
7182766	12/2006	Mogul	N/A	N/A
7229445	12/2006	Hayeck et al.	N/A	N/A
7241298	12/2006	Nemec et al.	N/A	N/A
7377924	12/2007	Raistrick et al.	N/A	N/A
7465303	12/2007	Riccione et al.	N/A	N/A
7641660	12/2009	Lakin et al.	N/A	N/A
D610257	12/2009	Horton	N/A	N/A
7686811	12/2009	Byrd et al.	N/A	N/A
D629900	12/2009	Fisher	N/A	N/A
7972338	12/2010	O'Brien	N/A	N/A
D646389	12/2010	Claypool et al.	N/A	N/A
8057478	12/2010	Kuczynski et al.	N/A	N/A
D651315	12/2010	Bertoni et al.	N/A	N/A
D651316	12/2010	May et al.	N/A	N/A
8080010	12/2010	Schulz et al.	N/A	N/A
8123753	12/2011	Poncet	N/A	N/A
8137406	12/2011	Novak et al.	N/A	N/A
8147530	12/2011	Strnad et al.	N/A	N/A
8167918	12/2011	Strnad et al.	N/A	N/A
8172848	12/2011	Tomko et al.	N/A	N/A
8177820	12/2011	Anapliotis et al.	N/A	N/A
8192441	12/2011	Collazo	N/A	N/A
8197487	12/2011	Poncet et al.	N/A	N/A
8231623	12/2011	Jordan	N/A	N/A
8231663	12/2011	Kay et al.	N/A	N/A
8246561	12/2011	Agee et al.	N/A	N/A
D666721	12/2011	Wright et al.	N/A	N/A
8262664	12/2011	Justin et al.	N/A	N/A
8282644	12/2011	Edwards	N/A	N/A

8282645	12/2011	Lawrence et al.	N/A	N/A
8292966	12/2011	Morton	N/A	N/A
8313492	12/2011	Wong et al.	N/A	N/A
8323289	12/2011	Re	N/A	N/A
8337503	12/2011	Lian	N/A	N/A
8343159	12/2012	Bennett	N/A	N/A
8377105	12/2012	Bscher	N/A	N/A
8382807	12/2012	Austin et al.	N/A	N/A
8403966	12/2012	Ralph et al.	N/A	N/A
D679395	12/2012	Wright et al.	N/A	N/A
8435246	12/2012	Fisher et al.	N/A	N/A
8475462	12/2012	Thomas et al.	N/A	N/A
8496690	12/2012	Sixto et al.	N/A	N/A
8523870	12/2012	Green, II et al.	N/A	N/A
D694884	12/2012	Mooradian et al.	N/A	N/A
D695402	12/2012	Dacosta et al.	N/A	N/A
8652142	12/2013	Geissler	N/A	N/A
D701303	12/2013	Cook	N/A	N/A
8672945	12/2013	Lavallee et al.	N/A	N/A
8696716	12/2013	Kartalian et al.	N/A	N/A
D705929	12/2013	Frey	N/A	N/A
8715363	12/2013	Ratron et al.	N/A	N/A
8728084	12/2013	Berelsman et al.	N/A	N/A
8758354	12/2013	Habegger et al.	N/A	N/A
8764763	12/2013	Wong et al.	N/A	N/A
8771279	12/2013	Philippon et al.	N/A	N/A
8784427	12/2013	Fallin et al.	N/A	N/A
8784457	12/2013	Graham	606/286	A61B 17/809
8795286	12/2013	Sand et al.	N/A	N/A
8801727	12/2013	Chan et al.	N/A	N/A
8808303	12/2013	Stemniski et al.	N/A	N/A
8828012	12/2013	May et al.	N/A	N/A
8828063	12/2013	Blitz	606/283	A61B 17/8085
8858602	12/2013	Weiner et al.	N/A	N/A
8882778	12/2013	Ranft	N/A	N/A
8888824	12/2013	Austin et al.	N/A	N/A
8911482	12/2013	Lee et al.	N/A	N/A
8940026	12/2014	Hilse et al.	N/A	N/A
8998903	12/2014	Price et al.	N/A	N/A
8998904	12/2014	Zeetser et al.	N/A	N/A
9023052	12/2014	Lietz et al.	N/A	N/A
9044250	12/2014	Olsen et al.	N/A	N/A
9060822	12/2014	Lewis	N/A	A61B 17/8061
9089376	12/2014	Medoff et al.	N/A	N/A
9101421	12/2014	Blacklidge	N/A	N/A
9107715	12/2014	Blitz et al.	N/A	N/A
9138244	12/2014	Mebarak et al.	N/A	N/A

9271769	12/2015	Batsch et al.	N/A	N/A
D765844	12/2015	DaCosta	N/A	N/A
D766434	12/2015	DaCosta	N/A	N/A
D766437	12/2015	DaCosta	N/A	N/A
D766438	12/2015	DaCosta	N/A	N/A
D766439	12/2015	DaCosta	N/A	N/A
9452057	12/2015	Dacosta et al.	N/A	N/A
9556946	12/2016	Bertram	N/A	F16H 55/18
9642656	12/2016	Kotuljac et al.	N/A	N/A
9668793	12/2016	Gaudin	N/A	N/A
9750538	12/2016	Soffiatti et al.	N/A	N/A
9867642	12/2017	Simon	N/A	N/A
9980760	12/2017	Dacosta et al.	N/A	N/A
10226287	12/2018	Langford et al.	N/A	N/A
10238437	12/2018	Simon	N/A	N/A
10376268	12/2018	Fallin et al.	N/A	N/A
11304705	12/2021	Fallin et al.	N/A	N/A
2002/0099381	12/2001	Maroney	N/A	N/A
2002/0107519	12/2001	Dixon et al.	N/A	N/A
2002/0198531	12/2001	Millard et al.	N/A	N/A
2003/0135212	12/2002	Chow	N/A	N/A
2004/0010259	12/2003	Keller et al.	N/A	N/A
2004/0039394	12/2003	Conti et al.	N/A	N/A
2004/0097946	12/2003	Dietzel et al.	N/A	N/A
2005/0004676	12/2004	Schon et al.	N/A	N/A
2005/0075641	12/2004	Singhatat et al.	N/A	N/A
2005/0101961	12/2004	Huebner et al.	N/A	N/A
2005/0149042	12/2004	Metzger	N/A	N/A
2005/0228389	12/2004	Stiernborg	N/A	N/A
2005/0267482	12/2004	Hyde, Jr.	N/A	N/A
2005/0273112	12/2004	McNamara	N/A	N/A
2006/0206044	12/2005	Simon	N/A	N/A
2006/0217733	12/2005	Plassky et al.	N/A	N/A
2006/0229621	12/2005	Cadmus	N/A	N/A
2006/0241607	12/2005	Myerson et al.	N/A	N/A
2006/0241608	12/2005	Myerson et al.	N/A	N/A
2006/0264961	12/2005	Murray-Brown	N/A	N/A
2007/0123857	12/2006	Deffenbaugh et al.	N/A	N/A
2007/0233138	12/2006	Figueroa et al.	N/A	N/A
2007/0265634	12/2006	Weinstein	N/A	N/A
2007/0276383	12/2006	Rayhack	N/A	N/A
2008/0091197	12/2007	Coughlin	N/A	N/A
2008/0140081	12/2007	Heavener et al.	N/A	N/A
2008/0172054	12/2007	Claypool et al.	N/A	N/A
2008/0208252	12/2007	Holmes	N/A	N/A
2008/0262500	12/2007	Collazo	N/A	N/A
2008/0269908	12/2007	Warburton	N/A	N/A
2009/0036893	12/2008	Kartalian et al.	N/A	N/A
2009/0093849	12/2008	Grabowski	N/A	N/A
2009/0105767	12/2008	Reiley	N/A	N/A

2009/0118733	12/2008	Orsak et al.	N/A	N/A
2009/0198244	12/2008	Leibel	N/A	N/A
2009/0198279	12/2008	Zhang et al.	N/A	N/A
2009/0222047	12/2008	Graham	N/A	N/A
2009/0254092	12/2008	Albiol Llorach	N/A	N/A
2009/0254126	12/2008	Orbay et al.	N/A	N/A
2009/0287309	12/2008	Walch et al.	N/A	N/A
2009/0312802	12/2008	Dasilva	N/A	N/A
2010/0069910	12/2009	Hasselman	N/A	N/A
2010/0121334	12/2009	Couture et al.	N/A	N/A
2010/0130981	12/2009	Richards	N/A	N/A
2010/0152782	12/2009	Stone et al.	N/A	N/A
2010/0168799	12/2009	Schumer	N/A	N/A
2010/0185245	12/2009	Paul et al.	N/A	N/A
2010/0249779	12/2009	Hotchkiss et al.	N/A	N/A
2010/0256687	12/2009	Neufeld	606/280	A61B 17/8061
2010/0324556	12/2009	Tyber et al.	N/A	N/A
2011/0087295	12/2010	Kubiak et al.	N/A	N/A
2011/0093084	12/2010	Morton	N/A	N/A
2011/0245835	12/2010	Dodds et al.	N/A	N/A
2011/0288550	12/2010	Orbay et al.	N/A	N/A
2011/0301648	12/2010	Lofthouse et al.	N/A	N/A
2012/0016426	12/2011	Robinson	N/A	N/A
2012/0065689	12/2011	Prasad et al.	N/A	N/A
2012/0078258	12/2011	Lo et al.	N/A	N/A
2012/0123420	12/2011	Honiball	N/A	N/A
2012/0123484	12/2011	Lietz et al.	N/A	N/A
2012/0130376	12/2011	Loring et al.	N/A	N/A
2012/0130383	12/2011	Budoff	N/A	N/A
2012/0184961	12/2011	Johannaber	N/A	N/A
2012/0239045	12/2011	Li	N/A	N/A
2012/0253350	12/2011	Anthony et al.	N/A	N/A
2012/0265301	12/2011	Demers et al.	N/A	N/A
2012/0277745	12/2011	Lizee et al.	N/A	N/A
2012/0303033	12/2011	Weiner et al.	N/A	N/A
2012/0330135	12/2011	Millahn et al.	N/A	N/A
2013/0012949	12/2012	Fallin et al.	N/A	N/A
2013/0035694	12/2012	Grimm et al.	N/A	N/A
2013/0085499	12/2012	Lian	N/A	N/A
2013/0096563	12/2012	Meade et al.	N/A	N/A
2013/0150900	12/2012	Haddad et al.	N/A	N/A
2013/0150903	12/2012	Vincent	N/A	N/A
2013/0158556	12/2012	Jones et al.	N/A	N/A
2013/0165936	12/2012	Myers	N/A	N/A
2013/0165938	12/2012	Chow et al.	N/A	N/A
2013/0172942	12/2012	Lewis et al.	N/A	N/A
2013/0184714	12/2012	Kaneyama et al.	N/A	N/A
2013/0190765	12/2012	Harris et al.	N/A	N/A
2013/0190766	12/2012	Harris et al.	N/A	N/A

2013/0204259	12/2012	Zajac	N/A	N/A
2013/0231668	12/2012	Olsen et al.	N/A	N/A
2013/0237987	12/2012	Graham	N/A	N/A
2013/0237989	12/2012	Bonutti	N/A	N/A
2013/0267956	12/2012	Terrill et al.	N/A	N/A
2013/0310836	12/2012	Raub et al.	N/A	N/A
2013/0325019	12/2012	Thomas et al.	N/A	N/A
2013/0325076	12/2012	Palmer et al.	N/A	N/A
2013/0331845	12/2012	Horan et al.	N/A	N/A
2013/0338785	12/2012	Wong	N/A	N/A
2014/0005672	12/2013	Edwards et al.	N/A	N/A
2014/0025127	12/2013	Richter	N/A	N/A
2014/0039501	12/2013	Schickendantz et al.	N/A	N/A
2014/0039561	12/2013	Weiner et al.	N/A	N/A
2014/0046387	12/2013	Waizenegger	N/A	N/A
2014/0074099	12/2013	Vigneron et al.	N/A	N/A
2014/0074101	12/2013	Collazo	N/A	N/A
2014/0094861	12/2013	Fallin	N/A	N/A
2014/0094924	12/2013	Hacking et al.	N/A	N/A
2014/0163563	12/2013	Reynolds et al.	N/A	N/A
2014/0171953	12/2013	Gonzalvez et al.	N/A	N/A
2014/0180342	12/2013	Lowery et al.	N/A	N/A
2014/0194884	12/2013	Martin et al.	N/A	N/A
2014/0207144	12/2013	Lee et al.	N/A	N/A
2014/0214037	12/2013	Mayer et al.	N/A	N/A
2014/0249537	12/2013	Wong et al.	N/A	N/A
2014/0257509	12/2013	Dacosta et al.	N/A	N/A
2014/0276815	12/2013	Riccione	N/A	N/A
2014/0276853	12/2013	Long et al.	N/A	N/A
2014/0277176	12/2013	Buchanan et al.	N/A	N/A
2014/0277214	12/2013	Helenbolt et al.	N/A	N/A
2014/0296995	12/2013	Reiley et al.	N/A	N/A
2014/0303621	12/2013	Gerold et al.	N/A	N/A
2014/0336658	12/2013	Luna et al.	N/A	N/A
2014/0350561	12/2013	Dacosta et al.	N/A	N/A
2015/0032168	12/2014	Orsak et al.	N/A	N/A
2015/0045801	12/2014	Axelsson, Jr. et al.	N/A	N/A
2015/0045839	12/2014	Dacosta et al.	N/A	N/A
2015/0051650	12/2014	Verstreken et al.	N/A	N/A
2015/0066094	12/2014	Prandi et al.	N/A	N/A
2015/0112446	12/2014	Melamed et al.	N/A	N/A
2015/0119944	12/2014	Geldwert	N/A	N/A
2015/0142064	12/2014	Perez et al.	N/A	N/A
2015/0150608	12/2014	Sammarco	N/A	N/A
2015/0182273	12/2014	Stemniski et al.	N/A	N/A
2015/0223851	12/2014	Hill et al.	N/A	N/A
2015/0245858	12/2014	Weiner et al.	N/A	N/A
2016/0015426	12/2015	Dayton et al.	N/A	N/A
2016/0022315	12/2015	Soffiatti et al.	N/A	N/A

2016/0135858	12/2015	Dacosta et al.	N/A	N/A
2016/0151165	12/2015	Fallin et al.	N/A	N/A
2016/0175089	12/2015	Fallin et al.	N/A	N/A
2016/0192950	12/2015	Dayton et al.	N/A	N/A
2016/0199076	12/2015	Fallin et al.	N/A	N/A
2016/0213384	12/2015	Fallin et al.	N/A	N/A
2016/0235414	12/2015	Hatch et al.	N/A	N/A
2016/0242791	12/2015	Fallin et al.	N/A	N/A
2016/0256204	12/2015	Patel et al.	N/A	N/A
2016/0354127	12/2015	Lundquist et al.	N/A	N/A
2017/0000533	12/2016	Fallin et al.	N/A	N/A
2017/0042599	12/2016	Bays et al.	N/A	N/A
2017/0079669	12/2016	Bays et al.	N/A	N/A
2018/0132868	12/2017	Dacosta et al.	N/A	N/A
2018/0344334	12/2017	Kim et al.	N/A	N/A
2018/0344371	12/2017	Monk et al.	N/A	N/A
2019/0357950	12/2018	Bernstein et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2009227957	12/2013	AU	N/A
2491824	12/2004	CA	N/A
2854997	12/2012	CA	N/A
695846	12/2005	CH	N/A
2930668	12/2006	CN	N/A
201558162	12/2009	CN	N/A
201572172	12/2009	CN	N/A
201586060	12/2009	CN	N/A
201912210	12/2010	CN	N/A
202801773	12/2012	CN	N/A
103462675	12/2012	CN	N/A
103505276	12/2013	CN	N/A
203458450	12/2013	CN	N/A
102860860	12/2013	CN	N/A
203576647	12/2013	CN	N/A
104490460	12/2014	CN	N/A
104510523	12/2014	CN	N/A
104523327	12/2014	CN	N/A
104546102	12/2014	CN	N/A
204379413	12/2014	CN	N/A
204410951	12/2014	CN	N/A
204428143	12/2014	CN	N/A
204428144	12/2014	CN	N/A
204428145	12/2014	CN	N/A
204446081	12/2014	CN	N/A
685206	12/1999	EP	N/A
1897509	12/2008	EP	N/A
2124772	12/2008	EP	N/A
2124832	12/2011	EP	N/A

2632349	12/2012	EP	N/A
2665428	12/2012	EP	N/A
2742878	12/2013	EP	N/A
2750617	12/2013	EP	N/A
2849684	12/2014	EP	N/A
3023068	12/2015	EP	N/A
2362616	12/1977	FR	N/A
2764183	12/1998	FR	N/A
3030221	12/2015	FR	N/A
2154143	12/1984	GB	N/A
2154144	12/1984	GB	N/A
200903719	12/2008	IN	N/A
200904479	12/2009	IN	N/A
140/DELNP/2012	12/2012	IN	N/A
2004/KOLNP/2013	12/2012	IN	N/A
4134243	12/2007	JP	N/A
4162380	12/2007	JP	N/A
2011092405	12/2010	JP	N/A
2011523889	12/2010	JP	N/A
4796943	12/2010	JP	N/A
5466647	12/2013	JP	N/A
2014511207	12/2013	JP	N/A
2014521384	12/2013	JP	N/A
5628875	12/2013	JP	N/A
100904142	12/2008	KR	N/A
756	12/2013	MD	N/A
2098036	12/1996	RU	N/A
2195892	12/2002	RU	N/A
2320287	12/2007	RU	N/A
2321366	12/2007	RU	N/A
2321369	12/2007	RU	N/A
2346663	12/2008	RU	N/A
2412662	12/2010	RU	N/A
1333328	12/1986	SU	N/A
0166022	12/2000	WO	N/A
2008051064	12/2007	WO	N/A
2009029798	12/2008	WO	N/A
2009032101	12/2008	WO	N/A
2011037885	12/2010	WO	N/A
2012029008	12/2011	WO	N/A
2013090392	12/2012	WO	N/A
2013134387	12/2012	WO	N/A
2013169475	12/2012	WO	N/A
2014020561	12/2013	WO	N/A
2014022055	12/2013	WO	N/A
2014035991	12/2013	WO	N/A
2014085882	12/2013	WO	N/A
2014147099	12/2013	WO	N/A
2014152219	12/2013	WO	N/A
2014152535	12/2013	WO	N/A

2014177783	12/2013	WO	N/A
2014200017	12/2013	WO	N/A
2015105880	12/2014	WO	N/A
2015127515	12/2014	WO	N/A

OTHER PUBLICATIONS

Dayton et al., "Dorsal Suspension Stitch: An Alternative Stabilization After Flexor Tenotomy for Flexible Hammer Digit Syndrome," *The Journal of Foot and Ankle Surgery*, vol. 48, No. 5, Sep./Oct. 2009, pp. 602-605. cited by applicant

Dayton et al., "The Extended Knee Hemilithotomy Position for Gastrocnemius Recession," *The Journal of Foot and Ankle Surgery*, vol. 49, 2010, pp. 214-216. cited by applicant

Wienke et al., "Bone Stimulation for Nonunions: What the Evidence Reveals," *Podiatry Today*, vol. 24, No. 9, Sep. 2011, pp. 52-57. cited by applicant

Dayton et al., "Hallux Varus as Complication of Foot Compartment Syndrome," *The Journal of Foot and Ankle Surgery*, vol. 50, 2011, pp. 504-506. cited by applicant

Dayton et al., "Measurement of Mid-Calcaneal Length on Plain Radiographs: Reliability of a New Method," *Foot and Ankle Specialist*, vol. 4, No. 5, Oct. 2011, pp. 280-283. cited by applicant

Dayton et al., "A User-Friendly Method of Pin Site Management for External Fixators," *Foot and Ankle Specialist*, Sep. 16, 2011, 4 pages. cited by applicant

Dayton et al., "Effectiveness of a Locking Plate in Preserving Midcalcaneal Length and Positional Outcome after Evans Calcaneal Osteotomy: A Retrospective Pilot Study," *The Journal of Foot and Ankle Surgery*, vol. 52, 2013, pp. 710-713. cited by applicant

Dayton et al., "Does Postoperative Showering or Bathing of a Surgical Site Increase the Incidence of Infection? A Systematic Review of the Literature," *The Journal of Foot and Ankle Surgery*, vol. 52, 2013, pp. 612-614. cited by applicant

Dayton et al., "Technique for Minimally Invasive Reduction of Calcaneal Fractures Using Small Bilateral External Fixation," *The Journal of Foot and Ankle Surgery*, Article in Press, 2014, 7 pages. cited by applicant

Dayton et al., "Clarification of the Anatomic Definition of the Bunion Deformity," *The Journal of Foot and Ankle Surgery*, vol. 53, 2014, pp. 160-163. cited by applicant

Dayton et al., "Observed Changes in Radiographic Measurements of the First Ray after Frontal Plane Rotation of the First Metatarsal in a Cadaveric Foot Model," *The Journal of Foot and Ankle Surgery*, Article in Press, 2014, 5 pages. cited by applicant

Dayton et al., "Observed Changes in First Metatarsal and Medial Cuneiform Positions after First Metatarsophalangeal Joint Arthrodesis," *The Journal of Foot and Ankle Surgery*, vol. 53, 2014, pp. 32-35. cited by applicant

Dayton et al., "Reduction of the Intermetatarsal Angle after First Metatarsal Phalangeal Joint Arthrodesis: A Systematic Review," *The Journal of Foot and Ankle Surgery*, Article in Press, 2014, 4 pages. cited by applicant

Feilmeier et al., "Reduction of Intermetatarsal Angle after First Metatarsophalangeal Joint Arthrodesis in Patients with Hallux Valgus," *The Journal of Foot and Ankle Surgery*, vol. 53, 2014, pp. 29-31. cited by applicant

Dayton et al., "Principles of Management of Growth Plate Fractures in the Foot and Ankle," *Clinics in Podiatric Medicine and Surgery, Pediatric Foot Deformities*, Oct. 2013, 17 pages. cited by applicant

Dayton et al., "Observed Changes in Radiographic Measurements of the First Ray after Frontal and Transverse Plane Rotation of the Hallux: Does the Hallux Drive the Metatarsal in a Bunion Deformity?," *The Journal of Foot and Ankle Surgery*, Article in Press, 2014, 4 pages. cited by applicant

Rodriguez et al., "Ilizarov method of fixation for the management of pilon and distal tibial fractures

in the compromised diabetic patient: A technique guide,” *The Foot and Ankle Journal Online*, vol. 7, No. 2, 2014, 9 pages. cited by applicant

Feilmeier et al., “Incidence of Surgical Site Infection in the Foot and Ankle with Early Exposure and Showering of Surgical Sites: A Prospective Observation,” *The Journal of Foot and Ankle Surgery*, vol. 53, 2014, pp. 173-175. cited by applicant

Catanese et al., “Measuring Sesamoid Position in Hallux Valgus: When Is the Sesamoid Axial View Necessary,” *Foot and Ankle Specialist*, 2014, 3 pages. cited by applicant

Dayton et al., “Comparison of Complications for Internal and External Fixation for Charcot Reconstruction: A Systematic Review,” *The Journal of Foot and Ankle Surgery*, Article in Press, 2015, 4 pages. cited by applicant

Dayton et al., “A new triplanar paradigm for bunion management,” *Lower Extremity Review*, Apr. 2015, 9 pages. cited by applicant

Dayton et al., “American College of Foot and Ankle Surgeons' Clinical Consensus Statement: Perioperative Prophylactic Antibiotic Use in Clean Elective Foot Surgery,” *The Journal of Foot and Ankle Surgery*, Article in Press, 2015, 7 pages. cited by applicant

Dayton et al., “Complications of Metatarsal Suture Techniques for Bunion Correction: A Systematic Review of the Literature,” *The Journal of Foot and Ankle Surgery*, Article in Press, 2015, 3 pages. cited by applicant

DeCarbo et al., “The Weil Osteotomy: A Refresher,” *Techniques in Foot and Ankle Surgery*, vol. 13, No. 4, Dec. 2014, pp. 191-198. cited by applicant

DeCarbo et al., “Resurfacing Interpositional Arthroplasty for Degenerative Joint Diseases of the First Metatarsalphalangeal Joint,” *Podiatry Management*, Jan. 2013, pp. 137-142. cited by applicant

DeCarbo et al., “Locking Plates: Do They Prevent Complications?,” *Podiatry Today*, Apr. 2014, 7 pages. cited by applicant

Easley et al., “Current Concepts Review: Hallux Valgus Part II: Operative Treatment,” *Foot and Ankle International*, vol. 28, No. 6, Jun. 2007, pp. 748-758. cited by applicant

Kim et al., “A Multicenter Retrospective Review of Outcomes for Arthrodesis, Hemi-Metallic Joint Implant, and Resectional Arthroplasty in the Surgical Treatment of End-Stage Hallux Rigidus,” *The Journal of Foot and Ankle Surgery*, vol. 51, 2012, pp. 50-56. cited by applicant

Easley et al., “Current Concepts Review: Hallux Valgus Part I: Pathomechanics, Clinical Assessment, and Nonoperative Management,” *Foot and Ankle International*, vol. 28, No. 5, May 2007, pp. 654-659. cited by applicant

Sandhu et al., “Digital Arthrodesis With a One-Piece Memory Nitinol Intramedullary Fixation Device: A Retrospective Review,” *Foot and Ankle Specialist*, vol. 6, No. 5, Oct. 2013, pp. 364-366. cited by applicant

Weber et al., “Use of the First Ray Splay Test to Assess Transverse Plane Instability Before First Metatarsocuneiform Fusion,” *The Journal of Foot and Ankle Surgery*, vol. 45, No. 4, Jul./Aug. 2006, pp. 278-282. cited by applicant

Smith et al., “Opening Wedge Osteotomies for Correction of Hallux Valgus: A Review of Wedge Plate Fixation,” *Foot and Ankle Specialist*, vol. 2, No. 6, Dec. 2009, pp. 277-282. cited by applicant

Easley et al., “What is the Best Treatment for Hallux Valgus?,” *Evidence-Based Orthopaedics—The Best Answers to Clinical Questions*, Chapter 73, 2009, pp. 479-491. cited by applicant

Shurnas et al., “Proximal Metatarsal Opening Wedge Osteotomy,” *Operative Techniques in Foot and Ankle Surgery*, Section I, Chapter 13, 2011, pp. 73-78. cited by applicant

Coetzee et al., “Revision Hallux Valgus Correction,” *Operative Techniques in Foot and Ankle Surgery*, Section I, Chapter 15, 2011, pp. 84-96. cited by applicant

Le et al., “Tarsometatarsal Arthrodesis,” *Operative Techniques in Foot and Ankle Surgery*, Section II, Chapter 40, 2011, pp. 281-285. cited by applicant

Collan et al., “The biomechanics of the first metatarsal bone in hallux valgus: A preliminary study utilizing a weight bearing extremity CT,” *Foot and Ankle Surgery*, vol. 19, 2013, pp. 155-161. cited

by applicant

Eustace et al., “Hallux valgus, first metatarsal pronation and collapse of the medial longitudinal arch—a radiological correlation,” *Skeletal Radiology*, vol. 23, 1994, pp. 191-194. cited by applicant

Mizuno et al., “Detorsion Osteotomy of the First Metatarsal Bone in Hallux Valgus,” *Japanese Orthopaedic Association*, Tokyo, 1956; 30:813-819. cited by applicant

Okuda et al., “The Shape of the Lateral Edge of the First Metatarsal Head as a Risk Factor for Recurrence of Hallux Valgus,” *The Journal of Bone and Joint Surgery*, vol. 89, 2007, pp. 2163-2172. cited by applicant

Okuda et al., “Proximal Metatarsal Osteotomy for Hallux Valgus: Comparison of Outcome for Moderate and Severe Deformities,” *Foot and Ankle International*, vol. 29, No. 7, Jul. 2008, pp. 664-670. cited by applicant

D'Amico et al., “Motion of the First Ray: Clarification Through Investigation,” *Journal of the American Podiatry Association*, vol. 69, No. 1, Jan. 1979, pp. 17-23. cited by applicant

Groves, “Operative Report,” St. Tammany Parish Hospital, Date of Procedure, Mar. 26, 2014, 2 pages. cited by applicant

Claim Chart for Groves Public Use (Mar. 26, 2014), Exhibit B4 of Defendant Fusion Orthopedics LLC's Invalidity Contentions, No. CV-22-00490-PHX-SRB, US District Court for the District of Arizona, Aug. 27, 2022, 161 pages. cited by applicant

Defendant Fusion Orthopedics LLC's Invalidity Contentions, No. CV-22-00490-PHX-SRB, US District Court for the District of Arizona, Aug. 27, 2022, 41 pages. cited by applicant

Prior Art Publications, Exhibit A of Defendant Fusion Orthopedics LLC's Invalidity Contentions, No. CV-22-00490-PHX-SRB, US District Court for the District of Arizona, Aug. 27, 2022, 3 pages. cited by applicant

Claim Chart for Fishco, “Making the Lapidus Easy,” *The Podiatry Institute* (Apr. 2014), Exhibit B1 of Defendant Fusion Orthopedics LLC's Invalidity Contentions, No. CV-22-00490-PHX-SRB, US District Court for the District of Arizona, Aug. 27, 2022, 97 pages. cited by applicant

Claim Chart for Fishco, “A Straightforward Guide to the Lapidus Bunionectomy,” *HMP Global* (Sep. 6, 2013), Exhibit B2 of Defendant Fusion Orthopedics LLC's Invalidity Contentions, No. CV-22-00490-PHX-SRB, US District Court for the District of Arizona, Aug. 27, 2022, 67 pages. cited by applicant

Claim Chart for Groves, “Functional Position Joint Sectioning: Pre-Load Method for Lapidus Arthrodesis,” *Update 2015: Proceedings of the Annual Meeting of the Podiatry Institute*, Chpt. 6, pp. 23-29 (Apr. 2015), Exhibit B3 of Defendant Fusion Orthopedics LLC's Invalidity Contentions, No. CV-22-00490-PHX-SRB, US District Court for the District of Arizona, Aug. 27, 2022, 151 pages. cited by applicant

Claim Chart for Mote, “First Metatarsal-Cuneiform Arthrodesis for the Treatment of First Ray Pathology: A Technical Guide,” *The Journal Foot & Ankle Surgery* (Sep. 1, 2009), Exhibit B5 of Defendant Fusion Orthopedics LLC's Invalidity Contentions, No. CV-22-00490-PHX-SRB, US District Court for the District of Arizona, Aug. 27, 2022, 21 pages. cited by applicant

Claim Chart for U.S. Pat. No. 10,376,268 to Fallin et al., entitled “Indexed Tri-Planar Osteotomy Guide and Method,” issued Aug. 13, 2019, Exhibit B6 of Defendant Fusion Orthopedics LLC's Invalidity Contentions, No. CV-22-00490-PHX-SRB, US District Court for the District of Arizona, Aug. 27, 2022, 155 pages. cited by applicant

Claim Chart for U.S. Pat. No. 8,282,645 to Lawrence et al., entitled “Metatarsal Bone Implant Cutting Guide,” issued Jan. 18, 2010, Exhibit B7 of Defendant Fusion Orthopedics LLC's Invalidity Contentions, No. CV-22-00490-PHX-SRB, US District Court for the District of Arizona, Aug. 27, 2022, 76 pages. cited by applicant

Claim Chart for U.S. Pat. No. 9,452,057 to Dacosta et al., entitled “Bone Implants and Cutting Apparatuses and Methods,” issued Apr. 8, 2011, Exhibit B8 of Defendant Fusion Orthopedics

LLC's Invalidity Contentions, No. CV-22-00490-PHX-SRB, US District Court for the District of Arizona, Aug. 27, 2022, 110 pages. cited by applicant

Obviousness Chart, Exhibit C of Defendant Fusion Orthopedics LLC's Invalidity Contentions, No. CV-22-00490-PHX-SRB, US District Court for the District of Arizona, Aug. 27, 2022, 153 pages. cited by applicant

"Foot and Ankle Instrument Set," Smith & Nephew, 2013, 2 pages. cited by applicant

"Lapidus Pearls: Gaining Joint Exposure to Decrease Non-Union," Youtube, Retrieved online from <<https://www.youtube.com/watch?v=-jqJyE7pj-Y>>, dated Nov. 2, 2009, 3 pages. cited by applicant

"Reconstructive Surgery of the Foot & Ankle," The Podiatry Institute, Update 2015, Conference Program, May 2015, 28 pages. cited by applicant

"Speed Continuous Active Compression Implant," BioMedical Enterprises, Inc., A120-029 Rev. 3, 2013, 4 pages. cited by applicant

"Visionaire: Patient Matched Cutting Blocks Surgical Procedure," Smith & Nephew, Inc., 2009, 2 pages. cited by applicant

Arthrex, "Comprehensive Foot System," Retrieved online from <https://www.arthrex.com/resources/animation/8U3iaPvY6kO8bwFAwZF50Q/comprehensive-foot-system?referringTeam=foot_and_ankle>, dated Aug. 27, 2013, 3 pages. cited by applicant

Baravarian, "Why the Lapidus Procedure is Ideal for Bunions," Podiatry Today, Retrieved online from <<https://www.hmpgloballearningnetwork.com/site/podipodi/article/5542>>, dated May 2006, 8 pages. cited by applicant

Bauer et al., "Offset-V Osteotomy of the First Metatarsal Shaft in Hallux Abducto Valgus," McGlamry's Comprehensive Textbook of Foot and Ankle Surgery, Fourth Edition, vol. 1, Chapter 29, 2013, 26 pages. cited by applicant

Cottom, "Fixation of the Lapidus Arthrodesis with a Plantar Interfragmentary Screw and Medial Low Profile Locking Plate," The Journal of Foot & Ankle Surgery, vol. 51, 2012, pp. 517-522. cited by applicant

Coughlin, "Fixation of the Lapidus Arthrodesis with a Plantar Interfragmentary Screw and Medial Low Profile Locking Plate," Orthopaedics and Traumatology, vol. 7, 1999, pp. 133-143. cited by applicant

Dayton et al., "Observed Changes in Radiographic Measurements of the First Ray after Frontal Plane Rotation of the First Metatarsal in a Cadaveric Foot Model," The Journal of Foot & Ankle Surgery, vol. 53, 2014, pp. 274-278. cited by applicant

Dayton et al., "Relationship of Frontal Plane Rotation of First Metatarsal to Proximal Articular Set Angle and Hallux Alignment in Patients Undergoing Tarsometatarsal Arthrodesis for Hallux Abducto Valgus: A Case Series and Critical Review of the Literature," The Journal of Foot & Ankle Surgery, 2013, Article in Press, Mar. 1, 2013, 7 pages. cited by applicant

DiDomenico et al., "Lapidus Bunionectomy: First Metatarsal-Cuneiform Arthrodesis," McGlamry's Comprehensive Textbook of Foot and Ankle Surgery, Fourth Edition, vol. 1, Chapter 31, 2013, 24 pages. cited by applicant

Fallin et al., US Provisional Application Entitled Indexed Tri-Planar Osteotomy Guide and Method, U.S. Appl. No. 62/118,378, filed Feb. 19, 2015, 62 pages. cited by applicant

Fishco, "A Straightforward Guide to the Lapidus Bunionectomy," Podiatry Today, Retrieved online from <<https://www.hmpgloballearningnetwork.com/site/podiatry/blogged/straightforward-guide-lapidus-bunionectomy>>, dated Sep. 6, 2013, 5 pages. cited by applicant

Fishco, "Making the Lapidus Easy," The Podiatry Institute, Update 2014, Chapter 14, 2014, pp. 91-93. cited by applicant

Fleming et al., "Results of Modified Lapidus Arthrodesis Procedure Using Medial Eminence as an Interpositional Autograft," The Journal of Foot & Ankle Surgery, vol. 50, 2011, pp. 272-275. cited by applicant

Fuhrmann, "Arthrodesis of the First Tarsometatarsal Joint for Correction of the Advanced

Splayfoot Accompanied by a Hallux Valgus,” *Operative Orthopadie und Traumatologie*, No. 2, 2005, pp. 195-210. cited by applicant

Gerard et al., “The Modified Lapidus Procedure,” *Orthopedics*, vol. 31, No. 3, Mar. 2008, 7 pages. cited by applicant

Giannoudis et al., “Hallux Valgus Correction,” *Practical Procedures in Elective Orthopaedic Surgery, Pelvis and Lower Extremity*, Chapter 38, 2012, 22 pages. cited by applicant

Greiner, “The Jargon of Pedal Movements,” *Foot & Ankle International*, vol. 28, No. 1, Jan. 2007, pp. 109-125. cited by applicant

Groves, “Functional Position Joint Sectioning: Pre-Load Method for Lapidus Arthrodesis,” *The Podiatry Institute, Update 2015*, Chapter 6, 2015, pp. 23-29. cited by applicant

Hardy et al., “Observations on Hallux Valgus,” *The Journal of Bone and Joint Surgery*, vol. 33B, No. 3, Aug. 1951, pp. 376-391. cited by applicant

Holmes, Jr., “Correction of the Intermetatarsal Angle Component of Hallux Valgus Using Fiberwire-Attached Endo-buttons,” *Revista Internacional de Ciencias Podologicas*, vol. 6, No. 2, 2012, pp. 73-79. cited by applicant

Integra, “Integra Large Qwix Positioning and Fixation Screw, Surgical Technique,” 2012, 16 pages. cited by applicant

Kilmartin et al., “Combined rotation scarf and Akin osteotomies for hallux valgus: a patient focused 9 year follow up of 50 patients,” *Journal of Foot and Ankle Research*, vol. 3, No. 2, 2010, 12 pages. cited by applicant

Lee et al., “Technique Tip: Lateral Soft-Tissue Release for Correction of Hallux Valgus Through a Medial Incision Using a Dorsal Flap Over the First Metatarsal,” *Foot & Ankle International*, vol. 28, No. 8, Aug. 2007, pp. 949-951. cited by applicant

Mote et al., “First Metatarsal-Cuneiform Arthrodesis for the Treatment of First Ray Pathology: A Technical Guide,” *JFAS Techniques Guide*, vol. 48, No. 5, Sep./Oct. 2009, pp. 593-601. cited by applicant

Myerson, “Cuneiform-Metatarsal Arthrodesis,” *The Foot and Ankle*, Chapter 9, 1997, pp. 107-117. cited by applicant

Sammarco, “Surgical Strategies: Mau Osteotomy for Correction of Moderate and Severe Hallux Valgus Deformity,” *Foot & Ankle International*, vol. 28, No. 7, Jul. 2007, pp. 857-864. cited by applicant

Schon et al., “Cuneiform-Metatarsal Arthrodesis for Hallux Valgus,” *The Foot and Ankle*, Second Edition, Chapter 8, 2002, pp. 99-117. cited by applicant

Sokoloff, “Lapidus Procedure,” *Textbook of Bunion Surgery*, Chapter 15, 1981, pp. 277-287. cited by applicant

Stamatis et al., “Mini Locking Plate as “Medial Buttress” for Oblique Osteotomy for Hallux Valgus,” *Foot & Ankle International*, vol. 31, No. 10, Oct. 2010, pp. 920-922. cited by applicant

Stewart, “Use for BME Speed Nitinol Staple Fixation for the Lapidus Procedure,” date unknown, 1 page. cited by applicant

Wukich et al., “Hypermobility of the First Tarsometatarsal Joint,” *Foot and Ankle Clinics*, vol. 10, No. 1, Mar. 2005, pp. 157-166. cited by applicant

Dayton et al., “Biwinged Excision for Round Pedal Lesions,” *The Journal of Foot and Ankle Surgery*, vol. 35, No. 3, 1996, pp. 244-249. cited by applicant

Dayton et al., “Medial Incision Approach to the First Metatarsophalangeal Joint,” *The Journal of Foot and Ankle Surgery*, vol. 40, No. 6, Nov./Dec. 2001, pp. 414-417. cited by applicant

Dayton et al., “Reduction of the Intermetatarsal Angle after First Metatarsophalangeal Joint Arthrodesis in Patients with Moderate and Severe Metatarsus Primus Adductus,” *The Journal of Foot and Ankle Surgery*, vol. 41, No. 5, Sep./Oct. 2002, pp. 316-319. cited by applicant

Dayton et al., “Use of the Z Osteotomy for Tailor Bunionectomy,” *The Journal of Foot and Ankle Surgery*, vol. 42, No. 3, May/Jun. 2003, pp. 167-169. cited by applicant

Dayton et al., "Early Weightbearing After First Metatarsophalangeal Joint Arthrodesis: A Retrospective Observational Case Analysis," *The Journal of Foot and Ankle Surgery*, vol. 43, No. 3, May/Jun. 2004, pp. 156-159. cited by applicant

Albano et al., "Biomechanical Study of Transcortical or Transtrabecular Bone Fixation of Patellar Tendon Graft with Bioabsorbable Pins in ACL Reconstruction in Sheep," *Revista Brasileira de Ortopedia (Rev Bras Ortop.)* vol. 47, No. 1, 2012, pp. 43-49. cited by applicant

Anderson et al., "Uncemented STAR Total Ankle Prostheses," *The Journal of Bone and Joint Surgery*, vol. 86(1, Suppl 2), Sep. 2004, pp. 103-111, (Abstract Only). cited by applicant

Dayton et al., "Is Our Current Paradigm for Evaluation and Management of the Bunion Deformity Flawed? A Discussion of Procedure Philosophy Relative to Anatomy," *The Journal of Foot and Ankle Surgery*, vol. 54, 2015, pp. 102-111. cited by applicant

Dayton et al., "Observed Changes in Radiographic Measurements of the First Ray after Frontal and Transverse Plane Rotation of the Hallux: Does the Hallux Drive the Metatarsal in a Bunion Deformity?," *The Journal of Foot and Ankle Surgery*, vol. 53, 2014, pp. 584-587. cited by applicant

Dayton et al., "Relationship of Frontal Plane Rotation of First Metatarsal to Proximal Articular Set Angle and Hallux Alignment in Patients Undergoing Tarsometatarsal Arthrodesis for Hallux Abducto Valgus: A Case Series and Critical Review of the Literature," *The Journal of Foot and Ankle Surgery*, vol. 52, No. 3, May/Jun. 2013, pp. 348-354. cited by applicant

Dayton et al., "Quantitative Analysis of the Degree of Frontal Rotation Required to Anatomically Align the First Metatarsal Phalangeal Joint During Modified Tarsal-Metatarsal Arthrodesis Without Capsular Balancing," *The Journal of Foot and Ankle Surgery*, 2015, pp. 1-6. cited by applicant

De Geer et al., "A New Measure of Tibial Sesamoid Position in Hallux Valgus in Relation to the Coronal Rotation of the First Metatarsal in CT Scans," *Foot and Ankle International*, Mar. 26, 2015, 9 pages. cited by applicant

DiDomenico et al., "Correction of Frontal Plane Rotation of Sesamoid Apparatus during the Lapidus Procedure: A Novel Approach," *The Journal of Foot and Ankle Surgery*, vol. 53, 2014, pp. 248-251. cited by applicant

Dobbe et al. "Patient-Tailored Plate for Bone Fixation and Accurate 3D Positioning in Corrective Osteotomy," *Medical and Biological Engineering and Computing*, vol. 51, No. 1-2, Feb. 2013, pp. 19-27, (Abstract Only). cited by applicant

EBI Extra Small Rail Fixator, Biomet Trauma, retrieved Dec. 19, 2014, from the Internet: <<http://footandanklefixation.com/product/biomet-trauma-ebi-extra-small-rail-fixator>>, 7 pages. cited by applicant

Garthwait, "Accu-Cut System Facilitates Enhanced Precision," *Podiatry Today*, vol. 18, No. 6, Jun. 2005, 6 pages. cited by applicant

Gonzalez Del Pino et al., "Variable Angle Locking Intercarpal Fusion System for Four-Corner Arthrodesis: Indications and Surgical Technique," *Journal of Wrist Surgery*, vol. 1, No. 1, Aug. 2012, pp. 73-78. cited by applicant

Grondal et al., "A Guide Plate for Accurate Positioning of First Metatarsophalangeal Joint during Fusion," *Operative Orthopädie Und Traumatologie*, vol. 16, No. 2, 2004, pp. 167-178 (Abstract Only). cited by applicant

"Hat-Trick Lesser Toe Repair System," Smith & Nephew, Brochure, Aug. 2014, 12 pages. cited by applicant

"Hoffmann II Compact External Fixation System," Stryker, Brochure, Literature No. 5075-1-500, 2006, 12 pages. cited by applicant

"Hoffmann II Micro Lengthener," Stryker, Operative Technique, Literature No. 5075-2-002, 2008, 12 pages. cited by applicant

"Hoffmann Small System External Fixator Orthopedic Instruments," Stryker, retrieved Dec. 19, 2014, from the Internet: <http://www.alibaba.com/product-detail/Stryker-Hoffmann-Small-System-External-Fixator_1438850129.html>, 3 pages. cited by applicant

Kim et al., "A New Measure of Tibial Sesamoid Position in Hallux Valgus in Relation to the Coronal Rotation of the First Metatarsal in CT Scans," *Foot and Ankle International*, vol. 36, No. 8, 2015, pp. 944-952. cited by applicant

"Lag Screw Target Bow," Stryker Leibinger GmbH & Co. KG, Germany 2004, 8 pages. cited by applicant

MAC (Multi Axial Correction) Fixation System, Biomet Trauma, retrieved Dec. 19, 2014, from the Internet: <<http://footandanklefixation.com/product/biomet-trauma-mac-multi-axial-correction-fixation-system>>, 7 pages. cited by applicant

Michelangelo Bunion System, Surgical Technique, Instratek Incorporated, publication date unknown, 4 pages. cited by applicant

Mini Joint Distractor, Arthrex, retrieved Dec. 19, 2014, from the Internet: <<http://www.arthrex.com/foot-ankle/mini-joint-distractor/products>>, 2 pages. cited by applicant

MiniRail System, Small Bone Innovations, Surgical Technique, 2010, 24 pages. cited by applicant

Modular Rail System: External Fixator, Smith & Nephew, Surgical Technique, 2013, 44 pages. cited by applicant

Monnich et al., "A Hand Guided Robotic Planning System for Laser Osteotomy in Surgery," *World Congress on Medical Physics and Biomedical Engineering* vol. 25/6: Surgery, Minimal Invasive Interventions, Endoscopy and Image Guided Therapy, Sep. 7-12, 2009, pp. 59-62, (Abstract Only). cited by applicant

Moore et al., "Effect of Ankle Flexion Angle on Axial Alignment of Total Ankle Replacement," *Foot and Ankle International*, vol. 31, No. 12, Dec. 2010, pp. 1093-1098, (Abstract Only). cited by applicant

Mortier et al., "Axial Rotation of the First Metatarsal Head in a Normal Population and Hallux Valgus Patients," *Orthopaedics and Traumatology: Surgery and Research*, vol. 98, 2012, pp. 677-683. cited by applicant

Okuda et al., "Postoperative Incomplete Reduction of the Sesamoids as a Risk Factor for Recurrence of Hallux Valgus," *The Journal of Bone and Joint Surgery*, vol. 91-A, No. 1, Jul. 2009, pp. 1637-1645. cited by applicant

Rx-Fix Mini Rail External Fixator, Wright Medical Technology, Brochure, Aug. 15, 2014, 2 pages. cited by applicant

Scanlan et al. "Technique Tip: Subtalar Joint Fusion Using a Parallel Guide and Double Screw Fixation," *The Journal of Foot and Ankle Surgery*, vol. 49, Issue 3, May-Jun. 2010, pp. 305-309, (Abstract Only). cited by applicant

Scranton Jr. et al, "Anatomic Variations in the First Ray: Part I. Anatomic Aspects Related to Bunion Surgery," *Clinical Orthopaedics and Related Research*, vol. 151, Sep. 1980, pp. 244-255. cited by applicant

Siddiqui et al. "Fixation of Metatarsal Fracture With Bone Plate in a Dromedary Heifer," *Open Veterinary Journal*, vol. 3, No. 1, 2013, pp. 17-20. cited by applicant

Sidekick Stealth Rearfoot Fixator, Wright Medical Technology, Surgical Technique, Dec. 2, 2013, 20 pages. cited by applicant

Simpson et al., "Computer-Assisted Distraction Osteogenesis by Ilizarov's Method," *International Journal of Medical Robots and Computer Assisted Surgery*, vol. 4, No. 4, Dec. 2008, pp. 310-320, (Abstract Only). cited by applicant

Small Bone External Fixation System, Acumed, Surgical Technique, Effective date Sep. 2014, 8 pages. cited by applicant

Stableloc External Fixation System, Acumed, Product Overview, Effective date Sep. 2015, 4 pages. cited by applicant

Stahl et al., "Derotation of Post-Traumatic Femoral Deformities by Closed Intramedullary Sawing," *Injury*, vol. 37, No. 2, Feb. 2006, pp. 145-151, (Abstract Only). cited by applicant

Talbot et al., "Assessing Sesamoid Subluxation: How Good is the AP Radiograph?," *Foot and Ankle*

International, vol. 19, No. 8, Aug. 1998, pp. 547-554. cited by applicant

TempFix Spanning the Ankle Joint Half Pin and Transfixing Pin Techniques, Biomet Orthopedics, Surgical Technique, 2012, 16 pages. cited by applicant

Weber et al., "A Simple System for Navigation of Bone Alignment Osteotomies of the Tibia," International Congress Series, vol. 1268, Jan. 2004, pp. 608-613, (Abstract Only). cited by applicant

Whipple et al., "Zimmer Herbert Whipple Bone Screw System: Surgical Techniques for Fixation of Scaphoid and Other Small Bone Fractures," Zimmer, 2003, 59 pages. cited by applicant

Yakacki et al. "Compression Forces of Internal and External Ankle Fixation Devices with Simulated Bone Resorption," Foot and Ankle International, vol. 31, No. 1, Jan. 2010, pp. 76-85, (Abstract Only). cited by applicant

Yasuda et al., "Proximal Supination Osteotomy of the First Metatarsal for Hallux Valgus," Foot and Ankle International, vol. 36, No. 6, Jun. 2015, pp. 696-704. cited by applicant

Chang et al., "Lapidus Arthrodesis: A Different Perspective," Journal of the American Podiatric Medical Association, vol. 84, No. 6, Jun. 1994, pp. 281-288. cited by applicant

Horton et al., "Deformity Correction and Arthrodesis of the Midfoot with a Medial Plate," Foot & Ankle, vol. 14, No. 9, Nov./Dec. 1993, pp. 493-499. cited by applicant

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Background/Summary

RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/897,321, filed Aug. 29, 2022, which is a continuation of U.S. patent application Ser. No. 16/877,159, filed May 18, 2020 and issued as U.S. Pat. No. 11,426,219, on Aug. 30, 2022, which is a continuation of U.S. patent application Ser. No. 15/148,774, filed May 6, 2016 and issued as U.S. Pat. No. 10,653,467, on May 19, 2020, which claims the benefit of U.S. Provisional Application Ser. No. 62/157,561, filed May 6, 2015. The entire contents of each of these applications are incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure relates generally to bone plate devices and methods for fixing bone using plate devices.

BACKGROUND

(2) Bones, such as the bones of a foot, may be anatomically misaligned. In certain circumstances, surgical intervention is required to correctly align the bones to reduce patient discomfort and improve patient quality of life. Surgical intervention may involve cutting one or more of the misaligned bones and then physically realigning the bones into an anatomically corrected position. A bone plate or multiple bone plates may be used to hold the bones in the anatomically corrected position, helping to prevent the bones from shifting back to their misaligned position.

SUMMARY

(3) In general, this disclosure is directed to bone fixation systems and techniques for fixating bones. In some examples, a bone plating system includes an intra-osseous support structure configured to be placed in an opening formed between adjacent bones. For example, during a tarsal-metatarsal fusion procedure in which a first metatarsal is realigned with respect to a second metatarsal, the intra-osseous support structure may be placed within the osseous tissue of the first metatarsal and the medial cuneiform, spanning the tarsal-metatarsal joint. An opening or groove may be formed in

the end of the first metatarsal facing the medial cuneiform and also in the end of the medial cuneiform facing the first metatarsal, providing cavities in which opposed ends of the intra-osseous support structure are inserted. One or more fasteners can be used to secure the intra-osseous support structure to the bones in which the fastener is inserted. For instance, in the example of a tarsal-metatarsal fusion procedure, a fastener may be inserted into the medial cuneiform (e.g., from the dorsal toward the plantar side), securing the intra-osseous support structure to the medial cuneiform. A second fastener can be inserted into the first metatarsal (e.g., from the dorsal toward the plantar side), securing the intra-osseous support structure to the first metatarsal.

(4) In some applications, a bone plate is also applied on exterior surfaces of the bone portions into which the intra-osseous support structure is inserted. For example, one or more flat or curved bone plates may be applied to exterior surfaces of bone portions containing the intra-osseous support structure. Depending on the configuration, the exterior bone plate(s) may be in compression while the intra-osseous support structure is in tensions under load, providing a balanced fixation system to effectively fixation opposed portions of bone.

(5) In one example, a bone plating system is described that includes a fastener having a length and an intra-osseous support structure. The example specifies that the intra-osseous support structure is configured to be placed in an opening formed in a first bone portion and a second bone portion and has an aperture to receive the fastener.

(6) In another example, a method of plating a bone is described. The method includes forming an opening in a first bone portion and a second bone portion and placing an intra-osseous support structure in the opening. The method further includes inserting a first fastener through the first bone portion and into the intra-osseous support structure and inserting a second fastener through the second bone portion and into the intra-osseous support structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a bone plating system in accordance with an embodiment of the invention.

(2) FIG. 2 is a perspective view of a bone plating system in accordance with an embodiment of the invention.

(3) FIG. 3 is a perspective view of a bone plating system in accordance with an embodiment of the invention.

(4) FIG. 4 is a perspective view of a bone plating system in accordance with an embodiment of the invention, with bone portions depicted as transparent.

(5) FIG. 5 is a top view of a bone plating system in accordance with an embodiment of the invention, with bone portions depicted as transparent.

(6) FIG. 6 is a perspective view of a bone plating system in accordance with an embodiment of the invention, with bone portions depicted as transparent.

(7) FIG. 7 is a perspective view of a bone plating system in accordance with an embodiment of the invention, with bone portions depicted as transparent.

(8) FIG. 8 is a perspective view of a bone plating system in accordance with an embodiment of the invention, with bone portions depicted as transparent.

(9) FIG. 9 is a side view of a bone plating system in accordance with an embodiment of the invention, with bone portions depicted as transparent.

(10) FIG. 10 is an end view of a bone plating system in accordance with an embodiment of the invention, with bone portions depicted as transparent.

(11) FIG. 11 is a side view of a bone plating system in accordance with an embodiment of the invention, with bone portions depicted as transparent.

(12) FIG. 12 is an end view of a bone plating system in accordance with an embodiment of the invention, with bone portions depicted as transparent.

(13) FIG. 13 is a top plan view of an intra-osseous support structure in accordance with an embodiment of the invention.

(14) FIG. 14 is a top plan view of an intra-osseous support structure in accordance with an embodiment of the invention.

(15) FIG. 15 is a perspective view of a bone plating system in accordance with an embodiment of the invention, with bone portions depicted as transparent.

(16) FIG. 16 is a perspective view of a bone plating system in accordance with an embodiment of the invention, with bone portions depicted as transparent.

DETAILED DESCRIPTION

(17) The following detailed description is exemplary in nature and is not intended to limit the scope, applicability, or configuration of the invention in any way. Rather, the following description provides some practical illustrations for implementing exemplary embodiments of the present invention. Examples of constructions, materials, dimensions, and manufacturing processes are provided for selected elements, and all other elements employ that which is known to those of ordinary skill in the field of the invention. Those skilled in the art will recognize that many of the noted examples have a variety of suitable alternatives.

(18) Embodiments of the invention include a bone plating system. Embodiments of the system can be useful for providing structural support to bones subject to a surgical procedure, such as a bone alignment, osteotomy, fracture repair, and/or fusion procedure. Such a procedure may be performed, for example, on bones (e.g., adjacent bones separated by a joint or different portions of a single bone separated by a fracture) in the foot or hand. In one example, the procedure can be performed to correct an alignment between a metatarsal (e.g., a first metatarsal) and a cuneiform (e.g., a first cuneiform), such as a bunion correction. An example of such a procedure is a lapidus procedure. In another example, the procedure can be performed by modifying an alignment of a metatarsal (e.g., a first metatarsal). An example of such a procedure is a basilar metatarsal osteotomy procedure.

(19) As shown in FIG. 1, embodiments of the bone plating system 10 include an intra-osseous support structure 20. As shown, the intra-osseous support structure can be adapted to be positioned intra-osseously. In such embodiments, the intra-osseous support structure is positioned within a thickness of a bone, such that both of its major surfaces face bone (e.g., cancellous bone). In the embodiment shown, the intra-osseous support structure 20 is adapted to be positioned within an opening 30 of a first bone portion 34 and a second bone portion 38, the opening leading to a cavity or void within the respective bone portions. As shown, in a bone portion having a dorsal surface 40 and a plantar surface 42, the opening can be formed closer to the plantar surface than the dorsal surface (e.g., between about one-half and two-thirds through the thickness of the bone). In certain embodiments, the opening crosses a centerline of one or both bone portions. In such embodiments, the bone defining the opening, and the intra-osseous support structure placed therein, will be in tension under load in situ. Such an intra-osseous support structure can be useful for providing structural support to bones subject to a surgical procedure.

(20) The intra-osseous support structure 20 can include any useful form. In some embodiments, the intra-osseous support structure has a first major surface, a second major surface, and a perimeter edge extending between the first major surface and the second major surface. In the embodiment shown in FIG. 1, the intra-osseous support structure is generally planar as are its first and second major surfaces. The major surfaces can be devoid of any protrusions. In certain embodiments, one or both of the major surfaces can include a surface treatment such as a texture. In some embodiments (not shown), the intra-osseous support structure can include a portion generally perpendicular to a first major surface. For example, the intra-osseous support structure can include a portion that extends from a side (e.g., a medial side) and bends or curves in an upward (e.g.,

dorsal) or a downward (e.g. plantar) direction around and/or in apposition to a cortical surface of a bone.

(21) In the embodiment shown in FIG. 1, the intra-osseous support structure **20** is connected to a bone portion by at least one fastener **50**. In some embodiments, the intra-osseous support structure has a first portion for placement in the first bone portion **34** and a second portion for placement in the second bone portion **38**, and at least one aperture (not shown in FIG. 1) to receive a respective fastener can be provided on each portion. In situ, the fastener can extend through a bone surface (e.g., a dorsal surface) and a portion of the thickness of a bone to the intra-osseously positioned support structure and be received into an aperture thereof.

(22) As shown in FIGS. 2 and 3, some embodiments of the bone plating system **10** can include a bone plate **60**. In the embodiment shown, the plate is adapted to be positioned on an outer surface of the first bone portion **34** and an outer surface of the second bone portion **38**. As shown, in a bone portion having a dorsal surface and a plantar surface, the dorsal surface, and the plate placed thereon, will be in compression under load in situ. Accordingly, some embodiments of the plating system include a plate in compression and an intra-osseous support structure in tension under load in situ. Such a plating system can be useful for providing structural support to bones subject to a surgical procedure.

(23) The bone plate **60** can include any suitable form. In some embodiments, the bone plate has a bone facing surface and a surface opposite the bone facing surface. In certain embodiments, such as the embodiment shown in FIG. 2, the bone plate **60** includes a generally planar member having generally planar surfaces. In other embodiments, such as the embodiment shown in FIG. 3, the bone plate **60** includes a curved shape (about and/or along its longitudinal axis). For example, the surface facing the bone can be concave and the opposite surface can be convex. In certain embodiments, the surface of the bone plate facing the bone may also have at least one protrusion to engage with the surface of the bone.

(24) In embodiments of the plating system having a plate **60**, the plate and intra-osseous support structure **20** can be connected to the bone and each other by the at least one fastener **50**. In such embodiments, the bone plate **60** can have at least one aperture **64**, **66** to receive respective fasteners **50**, **52**. In the embodiment shown, the bone plate has a first portion for placement on the first bone portion **34** and a second portion for placement on the second bone portion **38**. At least one aperture **64**, **66** for receiving a respective fastener **50**, **52** can be provided on each portion. Further, the intra-osseous support structure **20** can have at least one aperture (not shown in FIGS. 2 and 3) aligned to receive the respective fastener. In situ, the fastener can extend through the bone plate, a surface of the bone, and a portion of the thickness of a bone to the intra-osseously positioned support structure and be received into the aperture thereof. After final placement, in some embodiments, the bone plate and intra-osseous support structure will be generally parallel to each other.

(25) In some embodiments, the aperture in the intra-osseous support structure can include an attachment mechanism configured to engage a fastener. The fastener and attachment mechanism can include any structure suitable for engagement. In some embodiments, the fastener includes a screw, and the attachment mechanism includes a threaded aperture to receive and engage the screw. The attachment mechanism can include guides to facilitate alignment with the fasteners. In some embodiments, the fastener has a length that is less than the thickness of the bone. In certain embodiments, the fastener will have a length between about one-half of the thickness of the bone and the entire thickness of the bone. For example, the fastener can have a length of about two-thirds the thickness of the bone. In some embodiments, the fastener can extend through the aperture of the intra-osseous support structure (optionally engaging an attachment member thereof) and engage bone on one or both sides of the intra-osseous support structure.

(26) Any number of fasteners and respective intra-osseous support structure apertures can be provided. In the embodiment shown in FIGS. 1-3, the intra-osseous support structure **20** includes two apertures to receive two respective fasteners **50**, **52**. In the embodiment shown in FIGS. 4 and

5, the intra-osseous support structure **20** includes four apertures **70, 72, 74, 76** to receive four respective fasteners **50, 52, 54, 56**. In other embodiments, the intra-osseous support structure includes three apertures to receive three respective fasteners. In an embodiment of an intra-osseous support structure having a portion generally perpendicular to a first major surface, such a portion may include one or more apertures for receiving a fastener.

(27) The fasteners and respective apertures can be provided in any orientation. In some embodiments, such as the embodiments shown in FIGS. **4**, the apertures **70, 72, 74, 76** have a longitudinal axis perpendicular to a first major surface of the intra-osseous support structure **20**. In other embodiments, such as the embodiment shown in FIGS. **6** and **7**, the apertures **70, 72** have a longitudinal axis that intersects a first major surface of the intra-osseous support structure **20** at a skewed angle (e.g., an angle ranging from about 20 degrees to about 40 degrees from perpendicular). Further, in some embodiments, fasteners **50, 52** can extend from the bone surface and through the bone generally parallel to each other. In other embodiments, first and second fasteners can extend from the bone surface and through the bone at a skewed angle relative to each other. In embodiments of the plating system that include a plate, apertures **64, 66** can be configured such that first and second fasteners **50, 52** can extend from the bone plate **60** generally parallel to each other or at a skewed angle relative to each other.

(28) As shown in FIGS. **8-12**, some embodiments of the plating system can include an additional support that does not engage the intra-osseous support structure. Such an additional support can be useful for providing rotational stability to the plated bone portions. In the embodiment shown in FIG. **8**, the additional support includes a staple **100** having an end in each bone portion **34, 38**. In the example shown in FIGS. **9-10**, the additional support includes a pin **110** extending across the bone portions **34, 38** at an angle (e.g., about 45 degrees). In the embodiment shown, the pin **110** includes threads **120** on its distal portion to engage bone. It also includes notches **130** on its proximal portion. The pin can be broken at a desired notch after installation. In the embodiment shown in FIGS. **11** and **12**, two pins **110** are provided. As shown, the pins are inserted such that they extend across the bone portions **34, 38** in a crossing pattern, each at an angle (e.g., about 45 degrees).

(29) The plating system can also include features useful for placing the intra-osseous support structure. As shown in FIG. **3**, the intra-osseous support structure **20** can include a stop **150** on the perimeter edge that extends past the first major surface or the second major surface. In some embodiments, the stop is on a medial side of the intra-osseous support structure. In other embodiments, as shown in FIG. **13-15**, a tab **170** extending beyond the perimeter edge (e.g., on a medial side) can be provided. The tab can be useful for placing the intra-osseous support structure within the bone, and, in some embodiments, can be provided with break seam such that it can be easily removed from the intra-osseous support structure after placement. FIG. **16** shows in intra-osseous support structure **20** with the tab removed.

(30) The plating system can be used to join any bone portions. In one example, the first bone portion and the second bone portion are portions of a single bone separated by a fracture. As a further example, the first bone portion and the second bone portion are portions of a single bone separated by an osteotomy. As another example, the first bone portion and the second bone portion can be portions of two different bones separated by a joint, such as a cuneiform (e.g., medial cuneiform) and a metatarsal (e.g., first metatarsal). In the two-bone example, the intra-osseous support structure can be placed intra-osseously in the cuneiform and the metatarsal in an opening that spans the joint therebetween (e.g., tarsal-metatarsal joint). In such an embodiment, fasteners having a length less than the thickness of the cuneiform and metatarsal, respectively, can be used to connect the intra-osseous support structure to the bones. In embodiments of the plating system having a bone plate, the bone facing surface of the bone plate can be placed facing a dorsal surface of the cuneiform and a dorsal surface of the metatarsal, spanning a joint therebetween, and the fasteners can extend through apertures defined by the plate.

(31) Embodiments of the invention also include methods of plating a bone, such as with the embodiments of bone plating systems described herein. Note the order of steps as described is only exemplary unless otherwise indicated. In some embodiments, after preparing the surgical area, the method can include the step of forming an opening in a first bone portion and a second bone portion. The opening can be formed from a side of the bone. The opening can be formed generally parallel with a longitudinal axis of the bone, or may be formed at an angle with respect to such longitudinal axis such that it crosses the longitudinal axis. The opening can be formed, e.g., by a saw, drill, mill, box chisel, router, or the like.

(32) The method can also include the steps of placing an intra-osseous support structure in the opening and aligning it in a desired position. In some embodiments, the intra-osseous support structure can be placed generally parallel to a longitudinal axis of the bone (e.g., toward a tension side of the longitudinal axis. In other embodiments, the intra-osseous support structure can be placed at a skewed angle relative to the longitudinal axis of the bone, such that it crosses the longitudinal axis of the bone. In such embodiments, at least a portion of the intra-osseous support structure will reside on a tension side of the longitudinal axis and another portion will reside on a compression side of the longitudinal axis. The method can also include the steps of inserting a first fastener through a first bone portion and engaging the first fastener with the intra-osseous support structure, and inserting a second fastener through a second bone portion and engaging the second fastener with the intra-osseous support structure to secure the plating system to the bone. In some embodiments, the step of placing the intra-osseous support structure in the opening includes placing a stop in apposition to the first bone portion or the second bone portion. In embodiments of intra-osseous support structures having tabs, the method can also include removing the tab after placement of the support structure. Embodiments of the method can also include attaching an additional support structure to the first bone portion and the second bone portion.

(33) In some embodiments, the method can also include the step of forming a first hole in the first bone portion from the first surface and toward an opposite surface and forming a second hole in the second bone portion from the second surface and toward an opposite surface. The first and second holes and the opening can intersect. The first and second holes can be formed, for example, with hand-driven or powered drills. In such embodiments, the fasteners can be inserted through the holes to engage an intra-osseous support structure placed within the opening.

(34) Embodiments of the method also include placing a bone plate having a first portion in apposition to a first surface of a first bone portion and a second portion in apposition to a second surface of a second bone portion, the bone plate having a first aperture in the first portion and a second aperture in the second portion. The bone plate can be initially held in position by pins and/or protrusions. The fasteners can be inserted through apertures defined by the plate.

(35) Thus, embodiments of the invention are disclosed. Although the present invention has been described with reference to certain disclosed embodiments, the disclosed embodiments are presented for purposes of illustration and not limitation, and other embodiments of the invention are possible. One skilled in the art will appreciate that various changes, adaptations, and modifications may be made without departing from the spirit of the invention.

Claims

1. A method comprising: performing an osteotomy on a metatarsal to separate the metatarsal into a first metatarsal portion and a second metatarsal portion; forming an opening in the second metatarsal portion between a dorsal surface and a plantar surface the second metatarsal portion; forming an opening in the first metatarsal portion between the dorsal surface and the plantar surface of the first metatarsal portion; placing a first portion of an intra-osseous support structure in the opening in the first metatarsal portion; placing a second portion of the intra-osseous support structure in the opening in the second metatarsal portion, the second portion comprising a first

planar surface, a second planar surface, and an aperture extending through the second portion from the first planar surface to the second planar surface, wherein placing the second portion of the intra-osseous support structure in the opening comprises positioning the aperture facing the dorsal surface of the second metatarsal portion; and inserting a fastener through the dorsal surface of the second metatarsal portion, through a portion of the second metatarsal portion dorsal of the first planar surface, into the aperture of the second portion of the intra-osseous support structure, and through at least a portion of the second metatarsal portion plantar of the second planar surface.

2. The method of claim 1, wherein the fastener comprises a screw.
3. The method of claim 2, wherein the aperture comprises a threaded aperture, and inserting the screw into the aperture comprises inserting the screw into the threaded aperture.
4. The method of claim 1, wherein inserting the fastener through the portion of the second metatarsal portion plantar of the second planar surface comprises inserting the fastener through a thickness of the second metatarsal portion less than an entire thickness of the second metatarsal portion.
5. The method of claim 4, wherein the thickness of the second metatarsal portion is within a range from one-half to two-thirds of the entire thickness of the second metatarsal portion.
6. The method of claim 1, wherein the first portion of the intra-osseous support structure comprises an aperture, and further comprising inserting a second fastener into the aperture in the first portion of the intra-osseous support structure.
7. The method of claim 6, wherein inserting the second fastener into the aperture in the first portion of the intra-osseous support structure comprises inserting the second fastener through the dorsal surface of the first metatarsal portion.
8. The method of claim 1, wherein placing the first portion of the intra-osseous support structure in the opening in the first metatarsal portion and placing the second portion of the intra-osseous support structure in the opening in the second metatarsal portion comprises fusing the first metatarsal portion to the second metatarsal portion.
9. The method of claim 1, wherein the metatarsal is a first metatarsal.
10. The method of claim 1, further comprising adjusting an alignment of the first metatarsal portion relative to the second metatarsal portion.
11. The method of claim 1, wherein placing the second portion of the intra-osseous support structure in the opening comprises positioning the aperture facing the dorsal surface of the second metatarsal portion.
12. The method of claim 1, wherein inserting the fastener through the dorsal surface of the second metatarsal portion and into the aperture in the second portion of the intra-osseous support structure comprises inserting the fastener through the aperture toward the plantar surface the second metatarsal portion.
13. A method comprising: performing an osteotomy on a metatarsal to separate the metatarsal into a first metatarsal portion and a second metatarsal portion; placing a first portion of an intra-osseous support structure in the first metatarsal portion between a dorsal surface of the first metatarsal portion and a plantar surface of the first metatarsal portion; placing a second portion of the intra-osseous support structure in the second metatarsal portion between the dorsal surface of the second metatarsal portion and the plantar surface of the second metatarsal portion, the second portion comprising a first planar surface, a second planar surface, and an aperture extending through the second portion from the first planar surface to the second planar surface, wherein placing the second portion of the intra-osseous support structure in the second metatarsal portion comprises positioning the aperture facing the dorsal surface of the second metatarsal portion; and inserting a screw through the dorsal surface of the second metatarsal portion, through a portion of the second metatarsal portion dorsal of the first planar surface, into the aperture of the second portion of the intra-osseous support structure, and through at least a portion of the second metatarsal portion plantar of the second planar surface.

14. The method of claim 13, further comprising forming an opening in the second metatarsal portion between the dorsal surface and the plantar surface the second metatarsal portion.
 15. The method of claim 13, further comprising forming an opening in the first metatarsal portion between the dorsal surface and the plantar surface of the first metatarsal portion.
 16. The method of claim 13, wherein the metatarsal is a first metatarsal.
 17. The method of claim 13, further comprising adjusting an alignment of the first metatarsal portion relative to the second metatarsal portion.
 18. The method of claim 13, wherein placing the second portion of the intra-osseous support structure in the second metatarsal portion comprises positioning the aperture facing the dorsal surface of the second metatarsal portion.
 19. The method of claim 13, wherein inserting the screw through the dorsal surface of the second metatarsal portion and into the aperture in the second portion of the intra-osseous support structure comprises inserting the screw through the aperture toward the plantar surface the second metatarsal portion.
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